Geometric Sequences and Series

ANSWER KEY

Geometric sequences

1 Consider the sequence $3, 6, 12, 24, \ldots$

a State the common ratio. $r = 2$

b Write a formula for the nth term. $a_n = 3 \cdot 2^{n-1}$

c Find a_{10}. $a_{10} = 1536$

2 In a geometric sequence, $a_2 = 6$ and $a_5 = 48$. Find the common ratio and the first term.

$r^3 = \frac{48}{6} = 8$, so $r = 2$ and $a_1 = 3$.

Geometric series

3 Find the sum of the first 8 terms of $5 + 10 + 20 + \cdots$

$S_8 = 5\frac{2^8 - 1}{2 - 1} = 5(255) = 1275$.

4 Evaluate each infinite sum, or state that it diverges:

a $18 + 6 + 2 + \frac{2}{3} + \cdots$ $r = \frac{1}{3}$: $S = \frac{18}{1 - 1/3} = 27$.

b $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots$ $r = -\frac{1}{2}$: $S = \frac{1}{1 + 1/2} = \frac{2}{3}$.

c $2 + 3 + 4.5 + \cdots$ $r = 1.5 > 1$: diverges.

5 Write the repeating decimal $0.\overline{7}$ as a fraction using an infinite geometric series.

$0.7 + 0.07 + \cdots$ with $a = \frac{7}{10}$, $r = \frac{1}{10}$: $S = \frac{7/10}{9/10} = \frac{7}{9}$.

6 A ball is dropped from 8 m and rebounds to $\frac{3}{4}$ of its previous height on each bounce. Find the total vertical distance it travels (falling and rising) before coming to rest.

Total $= 8 + 2\left(8 \cdot \frac{3}{4} + 8 \cdot \left(\frac{3}{4}\right)^2 + \cdots\right) = 8 + 2\frac{6}{1 - 3/4} = 8 + 48 = 56$ m.